

~ REVIEW PAPER ~

OBJECTIVE → To study the various microeconomic aspects of Dolo 650, a commonly used painkiller (Paracetamol Supplement).

COLLECTED DATA:

The data^{is} collected from Mr. Khakan Satpathy, medicine distributor of Cuttack, Odisha.

→ Data of major shareholders of Paracetamol supplement in Cuttack region.
(TABLE-1)

NAME OF TABLET	MANUFACTURER	PRICE IN 2018 (FOR 10 TABS)	PRICE IN 2019 (FOR 10 TABS)
PACIMOL 650	INTIMA (IPCA)	₹ 11.71	₹ 12.00
DOLD 650	MICRO LABS LIMITED	₹ 9.88	₹ 10.00
DOLIPRANE 650	NIKOLAS	₹ 11.60	₹ 11.80
CALPOL 650	GLAXO SMITH KLINE PHARMACEUTICALS	₹ 8.73	₹ 9.49

→ Data of quantity demanded of Dolo 650 at different time of an year.
(TABLE-2)

FOR THE YEAR 2018	PRICE (IN ₹)	QUANTITY DEMANDED (APPROX)	
JANUARY	9.88	69500	
FEBRUARY	9.88	73700	* PRICES HERE ARE FOR A PACK OF 10 TABS.
MARCH	9.88	75800	
APRIL	9.88	77000	
MAY	9.88	79900	
JUNE	9.88	80000	
JULY	9.88	79600	
AUGUST	9.88	75300	
SEPTEMBER	9.88	74200	

OCTOBER	9.88	73300
NOVEMBER	9.88	72100
DECEMBER	9.88	67000

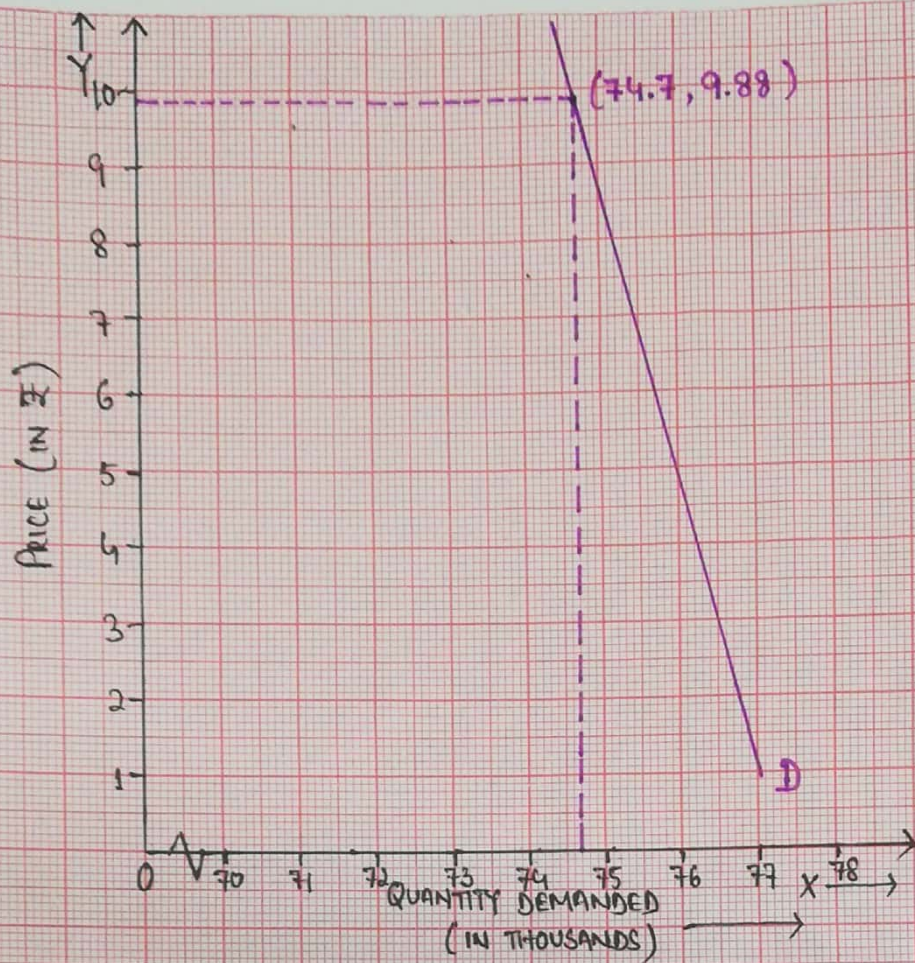
(TABLE-3)

FOR THE YEAR 2019	PRICE (IN ₹)	QUANTITY DEMANDED (APPROX)
JANUARY	10.00	68200
FEBRUARY	10.00	74500
MARCH	10.00	76000
APRIL	10.00	78100
MAY	10.00	80000
JUNE	10.00	82300
JULY	10.00	81700
AUGUST	10.00	77800
SEPTEMBER	10.00	75300

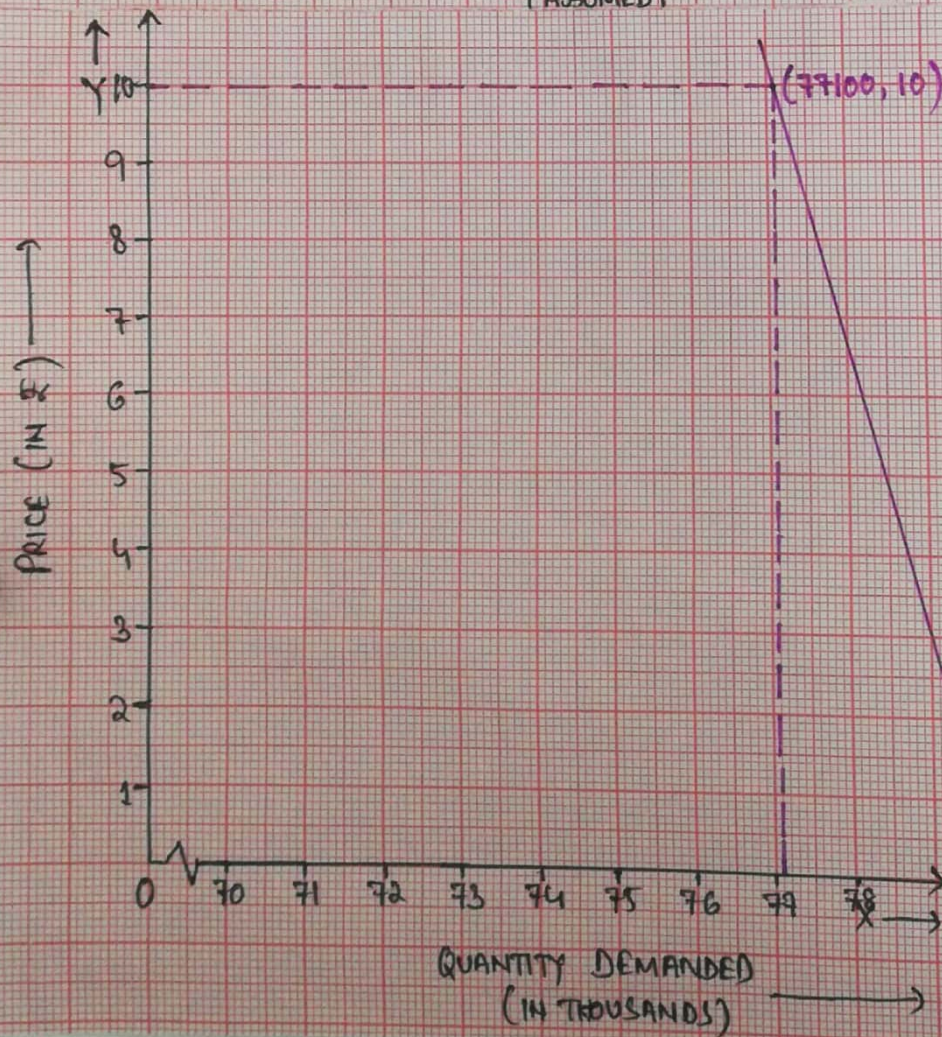
* PRICES HERE ARE FOR A PACK OF 10 TABS.

→ According to Mr. Khokan, Dolo 650 is the price maker in Cuttack region as all doctors prescribe it if generic medicines are not available in the Govt. supply. Henceforth, the prices of the medicines lie somewhere near to Dolo 650.

→ According to Dr. Satyjit Parhi, medicine specialist, MKCG Medical College and Hospital, the demand for Paracetamol supplements are higher during summers and early monsoons because this is when air-borne and water-borne diseases breed the highest and from August the human body gets adapted to the climate change and during the winters the rate of falling ill reduces so does the demand and then the demand starts increasing from early spring.



GRAPH-1: DEMAND FOR DOLO 650 IN 2018
(ASSUMED)



GRAPH-2: DEMAND FOR DOLO 650
2019
(ASSUMED)

→ According to Mr. Khokan, the medicines of same composition as that of Dolo 650 is available in Govt. Supply at ₹2.63 (GENERIC FORM) ^{for 10 TABS} whereas Dolo 650 costs ₹10.00 for 10 TABS. Because Dolo 650 is a product of Micro Labs Limited and it adds up its markup price and cost for brand value whereas the government charges only for the composition.

INTRODUCTION :-

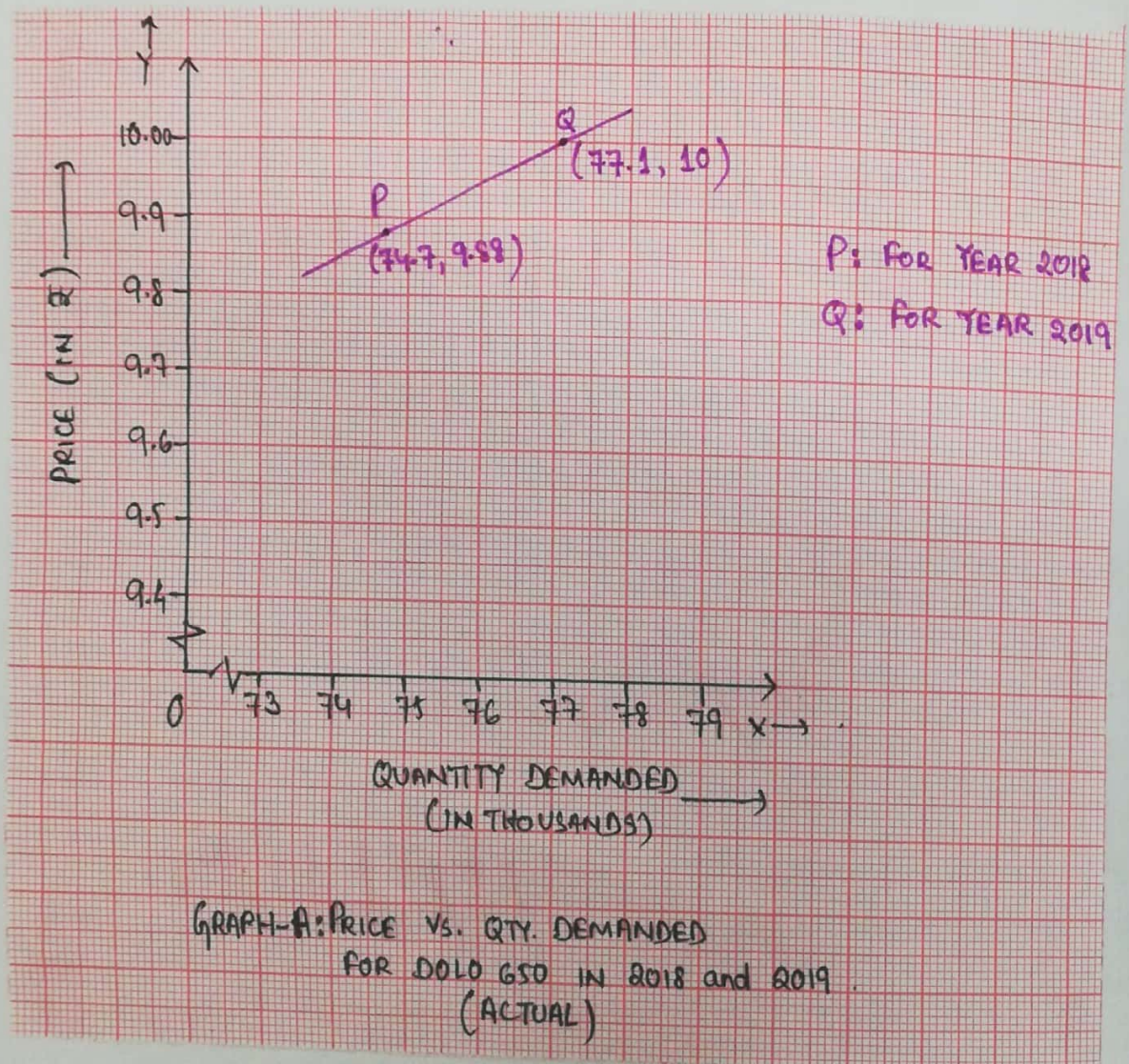
Paracetamol, also known as acetaminophen and APAP, is a medication used to treat pain and fever. It is typically used for mild to moderate pain relief. The dosage to be taken as per the consulted doctor or physician. It is not advisable to consume without a prescription. This is the most common painkiller composition in any allopathic medicinal tablets.

Whatever be a consumer's income or consumption he/she needs medicines when ill whatever its cost maybe and henceforth I have chosen this important commodity of our daily lives as a consumer does fall ill in each alternative day so medicine becomes a necessary good for a consumer at any point of year. And as paracetamol is the most common composition in medicines so finding data and writing a review becomes feasible also.

LAW OF DEMAND :-

The law of demand states that as the price of a commodity increases the demand for the commodity by a consumer decreases keeping other factors constant, *ceteris paribus* and vice versa.

According to TABLE-2 and TABLE-3, the price of Dolo 650 rises from ₹9.88 to ₹10.00 but the demand instead of falling from 2018 to 2019 increased as it is a necessary good hence it doesn't depend much on the price. So, the law of demand fails.



DEMAND CURVE :-

Demand curve is a graph showing how the demand for a commodity or service varies with changes in its price.

* GRAPH-A is the actual demand curve.

In Graph-1 and Graph-2 ~~even~~ if Dolo 650 earns abrupt profit as it is the price maker in Cuttack region but if it increases its price abruptly consumers will shift to other alternatives like,

- CALPOL 650
- DOLIPRANE 650
- PACIMOL 650

* Aggregate demand in 2018 is 74783.33 and in 2019 is 77100.

→ SUBSTITUTES :

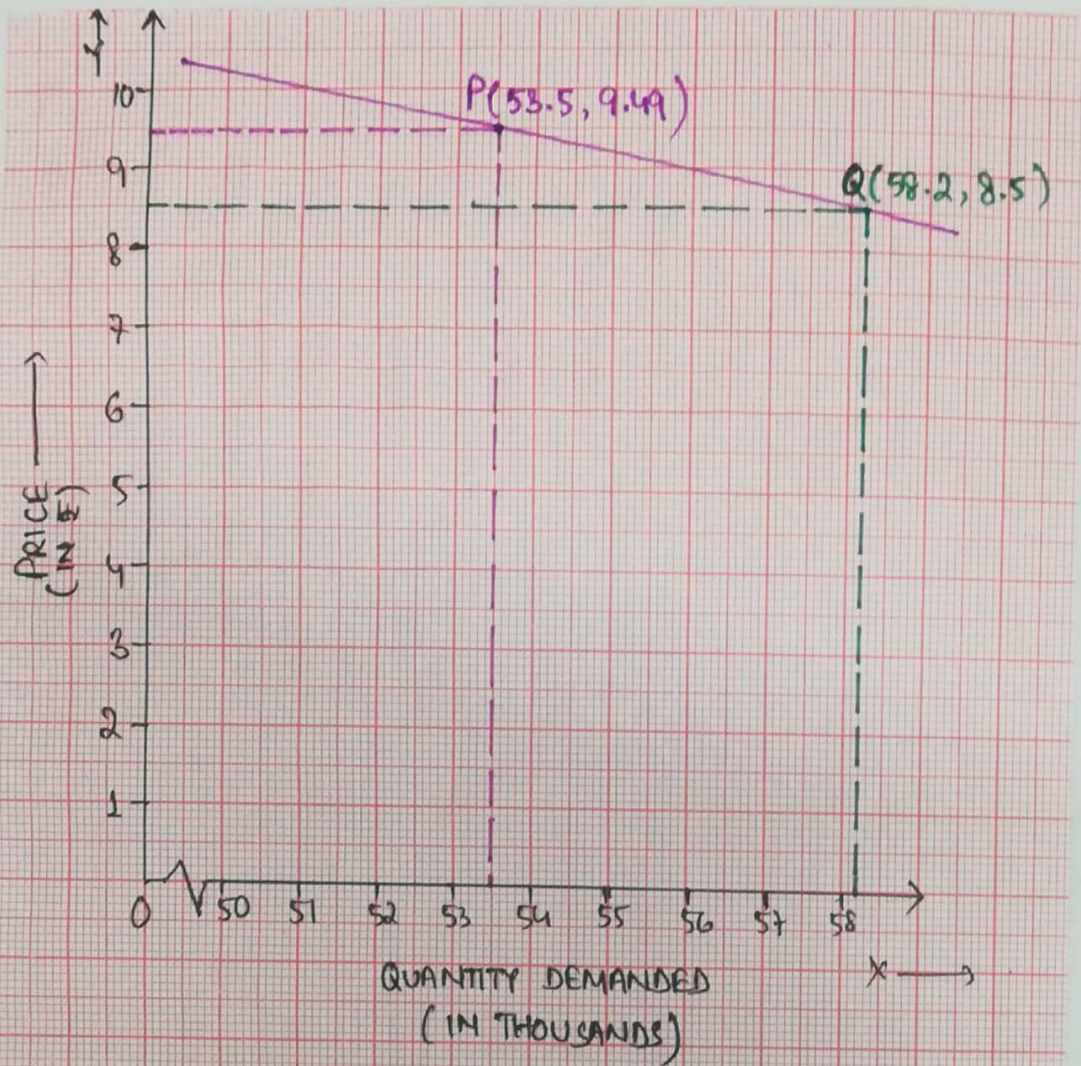
Goods that can serve as an alternative for one another, i.e., when the price of one good increases due to which the demand for the other increases and vice versa.

SUBSTITUTES FOR DOLO 650, (FOR 10 TABS)

- Calpol 650 at ₹9.49
- Doliprane 650 at ₹11.80
- Pacimol 650 at ₹12.00

But these substitutes do not have much demand as that of Dolo 650 in Cuttack region because Dolo 650 is prescribed by all doctors for its brand value and effectiveness. Most of the demand is fulfilled by Govt. Medicine Supply and the remaining demand is fulfilled by the private firm's supply.

The substitutes can't increase their demand by reducing the price to a small extent because the consumers trust Dolo 650's brand value and over years Dolo has gained this trust and good will. The substitutes have to decrease the price to a very low level which won't be suitable for the



GRAPH-3: DEMAND FOR CALPOL-650
IN 2019
(IMAGINARY)

* Even though shift from P to Q shows that with a less decrease in price the qty. demanded increases abruptly but this wouldn't happen in the market because Dolo 650 is the consumer preference because of its brand value and good will.

company's production unit to increase the demand for themselves as such a lower price would yield them losses in net.

→ COMPLIMENTARY GOODS :

Goods that 'go together', i.e., the decrease in price of a good results in increase in demand of its complement and vice versa.

As paracetamol is the only composition of its kind with different firms manufacturing it. There maybe other chemical composition with paracetamol for other purposes in Dolo 650 but it doesn't have a complement of itself.

TIME VS. DEMAND :-

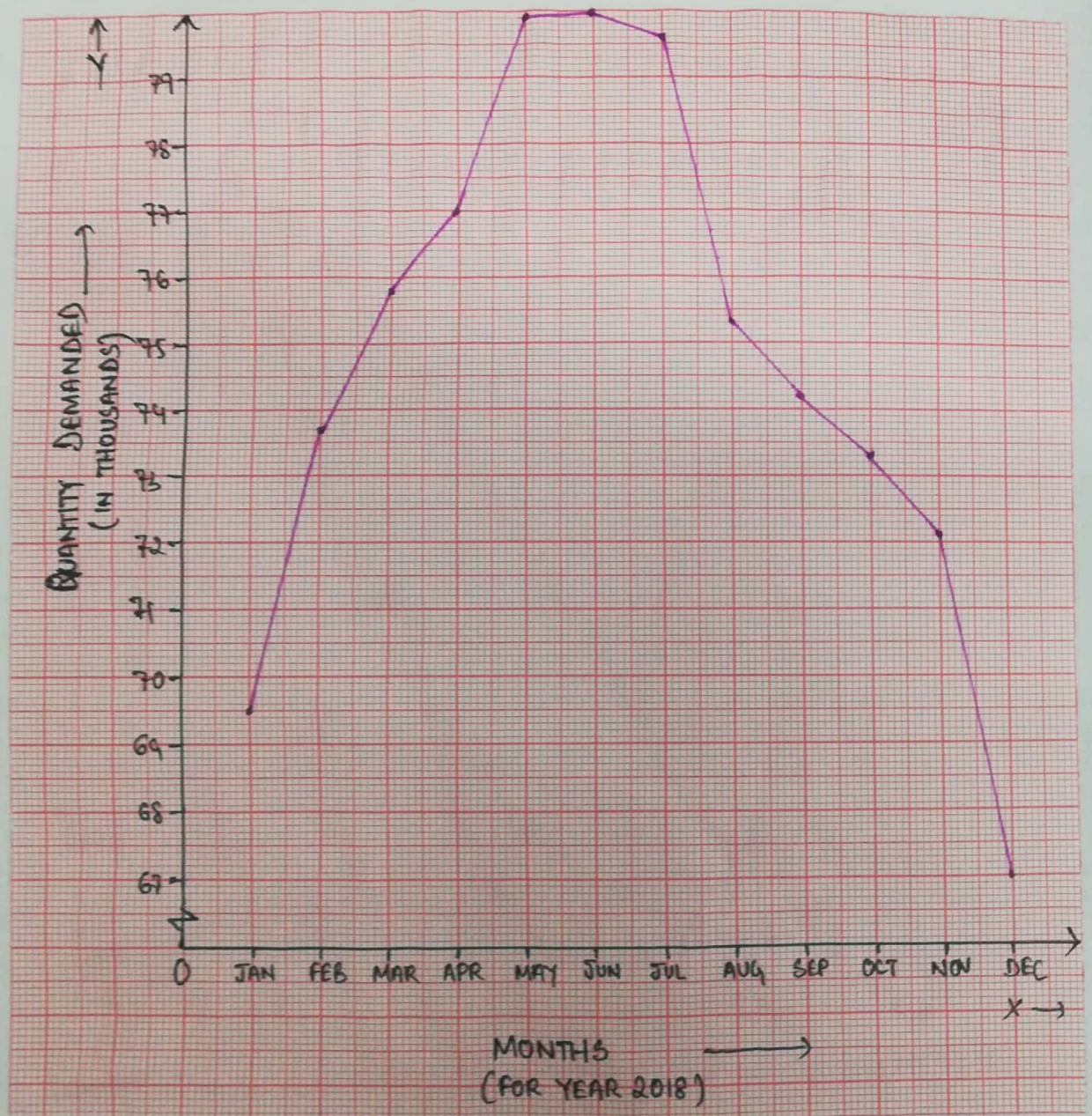
In Graph-4 and Graph-5 of TABLE-2 and TABLE-3 respectively shows the level of demand varies differently in different months of a year depending on the possibility to fall ill at different time in a year. The abrupt behaviour of demand curve is as per data collected from Mr. Khakan and the reason is due to explanation as given by Dr. Satyajit.

PRICE ELASTICITY OF DEMAND :-

It is a measure of how much the quantity demanded of a good responds to a change in the price of that good.

$$E_d = \frac{\Delta Q}{\Delta P} \times \frac{P_1 + P_2}{Q_1 + Q_2}$$

As per collected data, for Cuttack region in the year 2018 the avg. demand for Dolo 650 is 74783.33 and in the year 2019 the avg. demand for Dolo 650 is 77100. (as calculated from quantity demanded in TABLE-2 and TABLE-3)



GRAPH-4: DEMAND FOR DOLO 650 IN YEAR 2018.
(QTY. DEMANDED VS. MONTH)

$$E_d = \frac{Q_2 - Q_1}{P_2 - P_1} \times \frac{P_1 + P_2}{Q_1 + Q_2}$$

$$= \frac{77100 - 74783.83}{10 - 9.88} \times \frac{10 + 9.88}{77100 + 74783.88}$$

$$\approx 2.53 \text{ (APPROX.)}$$

As the qty. demanded increases with increase in price from 2018-19 so $E_d = +ve$
 E_d shows that Dolo 650 is highly elastic but over years as it has gained trust and good-will over consumers and, hence turned to be the price maker making it highly inelastic over other substitutes.

Even if Dolo 650 is highly inelastic and price maker of Cuttack region then also it can't increase the prices abruptly because this would lead to breaking of good-will among the consumers and shifting from Dolo 650 to some other substitutes.

COST ANALYSIS :-

It refers to monetary expenditure which a firm has to incur in order to purchase or hire the factors of production. It is the expense of purchasing or hiring factors of production and other business activities.

→ FIXED COSTS : Cost of fixed inputs used in production.

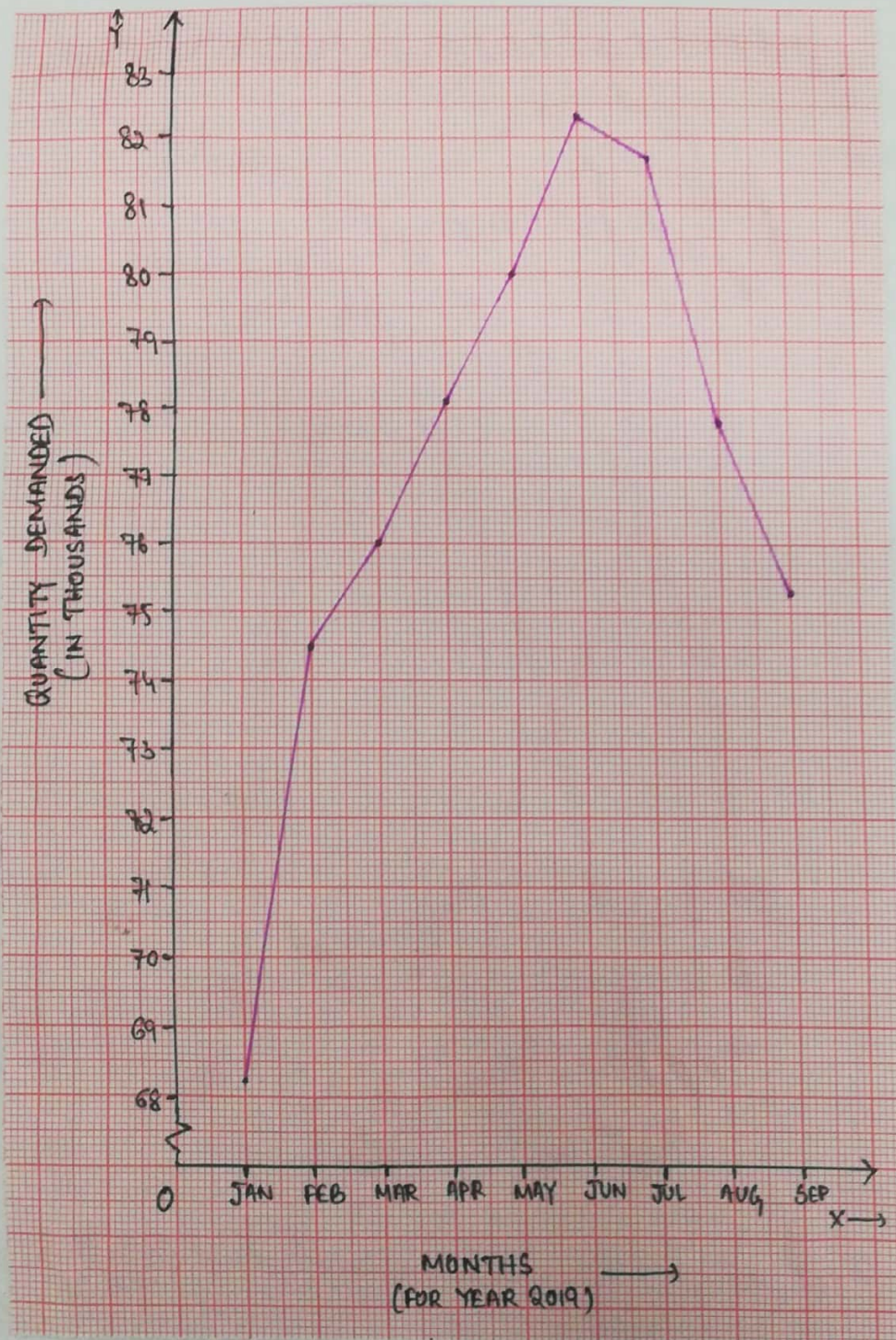
RABINDRA MEDICINE STORE

TYPE	COST / MONTH (IN ₹)
Godown RENT	3L
SALARIES	0.8L
MAINTAINANCE	1L

* DATA COLLECTED ARE
 ROUNDED OFF

Total fixed cost = ₹4.8 L per month

Data collected from Mr. Khokan Satapathy, medicine distributor from Cuttack, Odisha.



GRAPH-5 : DEMAND FOR DOLD 650 IN YEAR 2019
(QTY. DEMANDED VS. MONTHS)

→ VARIABLE COST : Cost of variable inputs used in production.

RABINDRA MEDICINE STORE

TYPE	COST / MONTH (IN ₹)
STOCKS	70L - 1CR.
ELECTRICITY	20K - 50K
MISC. EXPENSES	10K - 30K

* DATA COLLECTED ARE
ROUNDED OFF

Total Variable Costs = ₹70.3 L - ₹1.008 CR. per month

Data collected from Mr. Khokan Satpathy, medicine distributor from Cuttack, Odisha.

* TOTAL COSTS of Rabindra Medicine Store equals the sum of the total fixed costs and total variable costs for one month.

So, $TC = TVC + TFC$

$$\Rightarrow TC = ₹75.1 L - ₹1.056 CR. \text{ for 1 month}$$

SUNK COST :-

It is a cost that has already been incurred and cannot be recovered.

For Mr. Khokan, his sunk cost is the cost spent for getting the pharmaceutical license at ₹3.2 L.

MARKET STRUCTURE :-

Market structure refers to the number and types of firm in the industry.

For medicines, it comes under Oligopoly type of market structure as Oligopoly refers to a market situation in which there are few firms selling homogeneous or differentiated goods.

In Cuttack region, there are few firms who manufacture and sell paracetamol tablets with a good margin and Micro Lab. Ltd. being the best and popular so it turns to be the price maker for the sale of these tablets.

PRICE WAR :-

As it is an Oligopolistic Market, here price war is a competitive exchange among rival companies who lowered their price points on their products in a strategic attempt to ^{under}cut one another and capture greater market shares.

In Cuttack region, Dolo 650 has the more demand as it has created a good will among the consumers such that it has turned to be the price maker of the type and other company's tablets need to match its price and quality to have themselves in competition and have their shares in the market.

COMPETITORS :-

There are a lot of companies who make paracetamol tablets but the major shareholders in our region are Intima (IPCA), Nikolas, Glaxo Smith Kline Pharmaceuticals, and Micro Labs Ltd. The competition is high between these major companies but Micro Labs Ltd. is the price maker as it has created its good will among the consumers such that the change in its price also doesn't affect its demand and forces other firms to compete in terms of price and quality with Micro Lab. Ltd.

CONCLUSION :-

Dolo 650 does not follow the Law of Demand. It has mainly three substitutes except the Govt. Medicine Supply and no complimentary good. Even if the price elasticity of demand results to be approx. 2.53 which makes it a highly elastic good but due to the gained trust and good will it makes it a highly inelastic good. It is a price maker and other substitutes have to match up these prices or else their demand would crash. Dolo 650 has a different level of demand at different period of time in an year seeing the rate of diseases spreading at different months of an year. There does exist a price war between Dolo 650 and its substitutes but Dolo 650 earns abrupt profit because of consumer preference. This was an example of Oligopoly Market Structure. The generic composition of Dolo 650 costs around ₹2.63 per 10 TABS whereas cost of medicine manufactured by companies for same composition is high because of their higher markup price and brand value. In the price war of Dolo 650 the substitutes can't earn more demand by decreasing a lower rate of price, to earn more demand they have to cut down the prices abruptly squeezing their profit margin which in net turns to be loss for the firm. The demand for Dolo 650 increases with increase in price because from 2018 to 2019 the population increased so did the demand and as it is a necessary good price is not a factor of concern.

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